

Ethanol to Butanol Conversion over Bifunctional Zeotype Catalysts Containing Palladium and Zirconium

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Abstract—A study of the kinetics of ethanol conversion in the presence of Zr-containing zeolites BEA doped with palladium particles has revealed the order of formation of the main reaction products. It has been shown that the primary processes are ethanol dehydrogenation to acetaldehyde on Pd sites and ethanol dehydration to diethyl ether on the acid sites of the catalyst. After that, acetaldehyde undergoes the aldol–croton condensation reaction to form crotonal, which is hydrogenated to butanol on the metal sites. Butanol, in turn, is dehydrated into butenes, which undergo hydrogenation to butane. The presence of hydrogen in the gas phase leads to the displacement of ethanol from the metal surface and prevents the formation of surface carbonates and acetates. It has been found that hydrogen significantly accelerates ethanol dehydration owing to a decrease in the activation energy, which can be attributed to hydrogen spillover to the zeolite. The addition of water inhibits all acid-catalyzed reactions owing to competitive adsorption on acid sites and thereby decreases the butanol yield and the ethanol conversion.

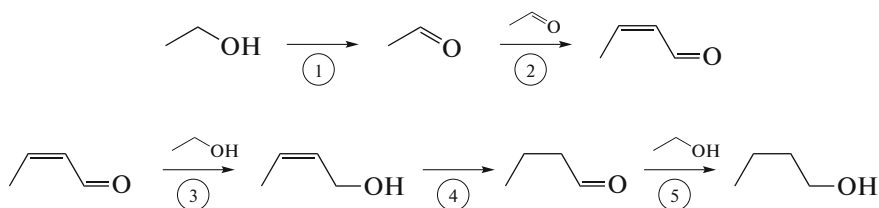
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At present, the catalytic conversion of ethanol to valuable chemicals is the focus of considerable attention in the scientific and patent literature [1]. This fact is associated with the tendency to redirecting the petrochemical industry to renewable raw materials. In recent decades, the global production of bioethanol from inedible biomass has increased; therefore, ethanol conversion is an alternative to the use of petroleum products. In the presence of solid acid catalysts, ethanol is converted to ethylene [2], butadiene, acetic acid, acetone, propylene [3], and isobutylene [4]. Many of these products are sought-after feedstocks in the production of elastomers, plastics, and other valuable materials. The global production of ethanol from biomass and cellulose is increasing annually; therefore, the problem of further ethanol conversion is extremely relevant in terms of sustainable development. Ethanol

conversion to butanol-1 is promising [5]. Unlike ethanol, butanol can be mixed with motor fuel in unlimited quantities; the energy characteristics of butanol are close to those of gasoline; therefore, butanol can be thought of as a full-fledged biofuel. In addition, butanol is commonly used in industry as a solvent, a perfumery component, and a perfume agent.

It is known that ethanol reacts with base catalysts to be converted to butanol and other higher alcohols $C_{2n}H_{2n+1}OH$ ($n = 2, 4$, etc.). For ethanol, this conversion, which is characteristic of other alcohols too, was first discovered by V.V. Markovnikov [6] and later studied by M. Guerbet [7]. Ethanol conversion to butanol can be described through a series of reactions (Scheme 1).



Scheme 1. Sequence of reactions in ethanol conversion to butanol according to Guerbet and Markovnikov [8, 9].

At the first stage (1), ethanol is converted to ethanal via a dehydrogenation reaction; after that, ethanal undergoes aldol–croton condensation (2) to form crotonal. The reduction of the carbonyl group in crotonal leads to the formation of crotyl alcohol (3), which undergoes isomerization to butanal (4). The second reduction of the carbonyl group leads to the formation of butanol (5) [10]. It is reported that the Guebet–Markovnikov process is implemented using bifunctional catalysts containing both noble metals (e.g., gold [11]) and an acid component (alumina), which is required for the occurrence of ethanal condensation. There are reports on catalysts Au/Al₂O₃ and Ni/Al₂O₃ [10] and bimetallic catalysts AuCu/Al₂O₃ [11]. Tests of a large number of metals dispersed on a support suggest that the activity of M/Al₂O₃ catalysts decreases in the following order: Ni > Pt > Au ~ Rh > Ru ≫ Ag. The best parameters (ethanol conversion of 25%, butanol selectivity of 80%) in the liquid-phase process are achieved at 250°C under a high pressure in the presence of a 20%Ni/Al₂O₃ catalyst. The authors of [9], in studying basic oxides, showed that palladium-doped MgAlO_x mixed oxides provide a high selectivity of ≥70% at 200°C in a closed system. The cited authors stated that Pd nanoparticles are the most optimum hydrogenation–dehydrogenation component of a bifunctional catalyst. It was found that, in the case of copper supported on mixed basic oxides in the Cu/MgAlO_x system, the strong base sites are active in butanol synthesis, while the acid sites mostly catalyze ethanol dehydration to ethylene and diethyl ether. However, Ho et al. [8] showed that calcium phosphate is an active catalyst for butanol synthesis. The authors of [12] showed that the selective conversion of ethanol to butanol requires the presence of optimum-strength acid sites in Ca₃(PO₄)₂. Thus, some acid catalysts can be active in the Guerbet process.

Lewis acid zeolites, in particular, zeolites of the BEA framework type, are commonly used as catalysts for a broad range of organic synthesis reactions [13–15]. Acid sites in the zeolite structure are formed via the isomorphous substitution of a silicon atom in the tetrahedral framework position by a Zr⁴⁺ [15] or Sn⁴⁺ atom [16], while the total electroneutrality of the framework is maintained. The sites formed in this way can catalyze stages 2–4 of ethanol conversion to butanol (Scheme 1). The doping of these molecular sieves with nanoparticles of a metal that is active in dehydrogenation/hydrogenation will also lead to an acceleration of stages 1 and 5 in the presence of a bifunctional catalyst. Similar metal-containing systems were used previously [9]; however, the base component of those catalysts was magnesium–aluminum mixed oxide.

This study is focused on ethanol conversion in the presence of a zirconium-containing Zr–BEA zeolite doped with palladium.

EXPERIMENTAL

Catalyst Synthesis Procedure

The zirconium-containing BEA zeolite was synthesized by hydrothermal synthesis in a fluoride-ion medium as described in [15]. The following gel composition was maintained constant: 1SiO₂ : 0.54TEAOH : 0.54HF : 0.005ZrO₂ : 5.6H₂O, (Si/Zr 200), where TEAOH is tetraethylammonium hydroxide. Tetraethoxysilane and zirconium oxychloride ZrOCl₂·8H₂O were used as a source of silicon and zirconium, respectively. Bifunctional catalysts were synthesized by incipient wetness impregnation. The original Zr–BEA zeolite powder was impregnated with a solution with a given concentration of Pd(NO₃)₂ in ethanol. After impregnation, the samples were dried in air for 2 h and then at 100°C for 10 h; after that, they were calcined in an air stream at 300°C for 5 h to decompose the nitrate.

Instrumental Investigation Procedures

X-ray diffraction (XRD) analysis data were obtained on a Bruker D2 Phaser instrument. Crystal size and morphology were studied by scanning electron microscopy (SEM) on a Hitachi TM3030 microscope at an accelerating voltage of 5 kV. Palladium particle size distribution was studied by transmission electron microscopy (TEM) on an FEI Osiris instrument (source with a field emission of 200 keV); images of about 100 particles were used for analysis. The pore structure of the catalysts was characterized by low-temperature nitrogen sorption. Isotherms were recorded at –196°C on a Micromeritics ASAP 2000 instrument after pre-evacuation at 350°C in a vacuum. Surface compounds were studied by IR spectroscopy. Infrared spectra were recorded in the transmission mode on a Thermo Scientific Nicolet iS-10 Fourier transform IR spectrometer equipped with an MCT detector cooled with liquid nitrogen. The spectra were recorded at a resolution of 4 cm^{–1} in a range of 4000–650 cm^{–1}. Before recording, the sample was compressed into a disk without a binder and placed in a special quartz cell with KBr windows, which was connected to a vacuum line. The sample was heated in a separate portion of the cell at 400°C for 1 h to a vacuum of 10^{–5} torr; after that, the spectrum of the pure sample was recorded. After ethanol adsorption, the spectra were subtracted from the spectrum of the pure sample. The sample was heated to a required temperature (50–250°C), held for 15 min, and then cooled to 25°C to record a spectrum.

Temperature-programmed desorption (TPD) of ethanol was studied in a quartz microreactor (internal diameter of 3 mm). A 0.6%Pd/Zr–BEA sample in the form of a fraction of 0.25–0.5 mm was placed between quartz wool layers and heated in a helium stream at 300°C for 1 h. After that, the sample was cooled to 60°C and saturated with ethanol fed through a syringe

dispenser. Next, weakly sorbed ethanol was blown off for 2 h; the sample was heated to 500°C at a rate of 10 deg/min. Desorbed compounds were recorded on-line at the exit of the reactor using a mass-selective detector.

Catalytic Testing Procedure

Ethanol conversion was studied in a flow reactor at atmospheric pressure and an ethanol feed space velocity of 0.15–6.5 g_{ethanol}/(g_{cat} h). The catalyst in the form of a fraction of 0.5–1 mm was placed in a quartz reactor and preconditioned in a nitrogen stream at 300°C for 1 h; next, it was cooled to 200°C, and hydrogen was fed for 1 h to reduce palladium particles; after that, the temperature was set at a required level. The tests were conducted at a constant stream of nitrogen or hydrogen mixed with nitrogen.

RESULTS AND DISCUSSION

Physicochemical Properties of the Catalysts

According to chemical analysis, the Zr–BEA zeolite sample prepared by hydrothermal synthesis had a Si/Zr molar ratio of 200 ± 20 , which is close to a value set during synthesis. According to low-temperature nitrogen adsorption, the sample is a microporous material with a pore volume of 0.25 cm³/g, which corresponds to highly crystalline zeolite BEA [15]. According to XRD (Fig. 1), the sample comprises only the BEA zeolite phase with a high degree of crystallinity. According to SEM, the zeolite crystal size is about 2–3 μm, which is characteristic of zeolites synthesized in an anion fluoride medium.

The state of palladium in the samples was studied by TEM; the results are shown in Fig. 2. The image shows the BEA zeolite crystallographic faces with an interplanar spacing of about 0.6 nm. Palladium particles (mostly of a spherical shape) with a diameter of about 3–4 nm are localized near the zeolite crystal; in addition, it was found that the sample contains larger palladium particles localized separately from the zeolite crystals.

The Pd particle size distribution shows that approximately 50% of the particles have a diameter of 3–6 nm. These particles are localized on the outer surface of the zeolite crystals. The other 50% of the palladium particles have a diameter larger than 10 nm and are localized separately in the form of aggregates.

Ethanol Conversion Products and Order of Their Formation

In the study of ethanol conversion in the presence of a catalyst containing 1 wt % Pd, the following reaction products were found: ethane, butane, acetaldehyde, diethyl ether, butanol, and ethyl butyl ether; in addition, traces of butanal, butenes, and ethyl acetate

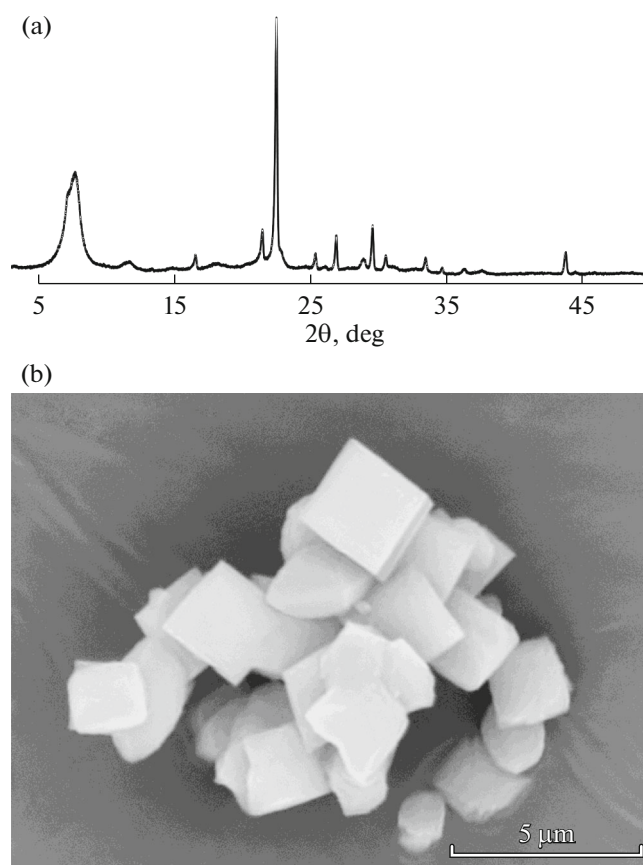


Fig. 1. (a) Diffraction pattern and (b) SEM image of the Zr–BEA sample.

were detected. To determine the order of formation of the products, curves of the dependence of the *i*th product yield on ethanol conversion were plotted (Fig. 3).

The dependence of the diethyl ether yield on conversion corresponds to the primary stable product. Diethyl ether is formed during the bimolecular dehydration of ethanol; it is not involved in secondary processes under these conditions. The curve of the acetaldehyde yield also corresponds to the primary product, the selectivity for which (i.e., the slope of the curve in Fig. 3) decreases with increasing conversion. Acetaldehyde is formed in the ethanol dehydrogenation reaction and then undergoes several secondary processes, the main of which is condensation to crotonal. Products of the aldol–croton condensation of acetaldehyde were not detected in the catalyzate; this fact is associated with the vigorous hydrogenation of the C=C and C=O bonds. With an increase in conversion, synchronously with a decrease in acetaldehyde selectivity, butanol and butane appear in the products. Apparently, at a short contact time, butanol and butane are formed with comparable selectivities; however, with an increase in conversion, the contribution of butanol conversion to butane increases. This find-

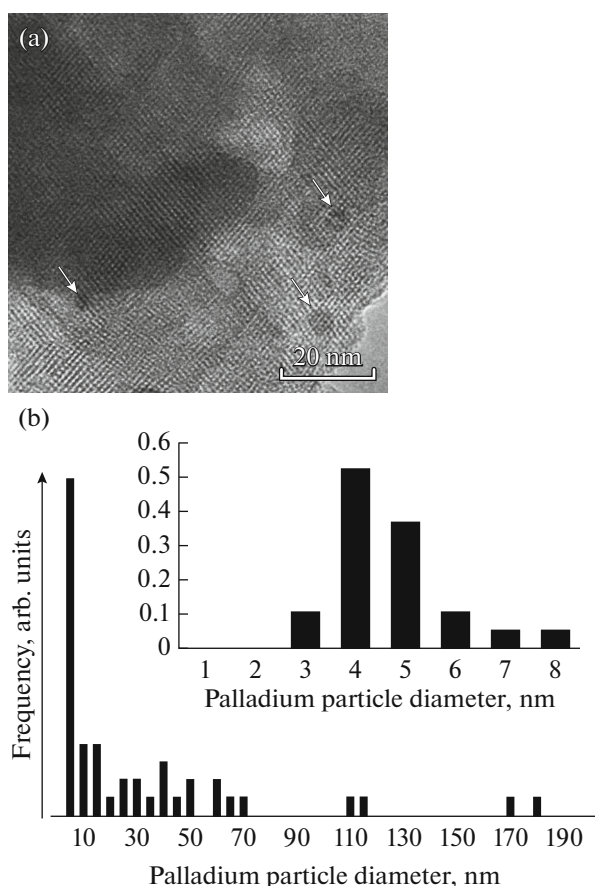


Fig. 2. (a) Representative TEM image (Pd particles are marked with arrows) and (b) Pd particle size distribution for the 0.6%Pd/Zr–BEA sample according to TEM.

ing can be attributed to butanol dehydration to butene on the acid sites of the catalyst and the subsequent butene hydrogenation to butane on the metal sites. The formation of butane at a low conversion can indicate that crotonal can undergo direct hydrogenation over Pd particles to hydrolyze the C–O bond. The curve of the ethane yield corresponds to the secondary product; this fact suggests that ethane is formed via the intermediate monomolecular dehydration of ethanol to ethylene, rather than from ethanol. The overall reaction scheme is shown in Fig. 4.

It should be noted that, under the process conditions, the hydrogenation reaction equilibrium is significantly shifted toward saturated hydrocarbons. A key requirement for acid sites of a bifunctional catalyst in this order of conversions is that they should exhibit high activity in acetaldehyde condensation and low activity in the dehydration of ethanol and butanol.

Surface Compounds According to IR Data

The structure and concentration of surface compounds formed in the reaction of ethanol and a Pd/Zr–BEA catalyst were studied by IR spectroscopy.

The sample was saturated with ethanol vapors at 25°C and then heated to 250°C with a stepwise increase in temperature; IR spectra of the surface compounds were recorded after cooling to 25°C (Fig. 5).

During ethanol adsorption, a “negative” band at 3732 cm^{-1} appears in the spectra owing to the formation of a hydrogen bond between the OH group of ethanol and the Si–OH $\cdots$ O(H)–C₂H₅ silanol group. The broad band at $\sim 3400\text{ cm}^{-1}$ corresponds to the same complex. In the region of the C–H stretching vibrations, the spectrum exhibits bands at 2980 ($\nu_{\text{as}}(\text{CH}_3)$), 2937 ($\nu_{\text{as}}(\text{CH}_2)$), 2904 ($\nu_{\text{s}}(\text{CH}_3)$), and 2880 cm^{-1} ($\nu_{\text{s}}(\text{CH}_2)$), which are attributed to the ethyl group of ethanol molecules [17]. In the region of bending vibrations, bands characteristic of ethanol are located at 1483 ($\delta_{\text{as}}(\text{CH}_2)$), 1451 ($\delta_{\text{as}}(\text{CH}_3)$), and 1395 cm^{-1} ($\delta_{\text{s}}(\text{CH})$). Upon heating to 50°C, the spectra exhibit a band of the carbonyl group vibrations $\nu(\text{C}=\text{O})$ at 1724 cm^{-1} and less intense broad bands at 1686 and 1641 cm^{-1} . This finding indicates the occurrence of dehydrogenation reactions to form acetaldehyde, which in turn, even at 50°C, can undergo the aldol–croton condensation reaction to form α,β -unsaturated carbonyl compounds. In addition, a broad band at 1619 cm^{-1} corresponds to bicarbonates $[(\text{HO})\text{CO}_2]^-$, which are also formed at a low temperature [18]. Further heating leads to an increase in the intensity of the bands corresponding to acetaldehyde. At 150°C, the spectrum exhibits intense bands at 1754 ($\nu(\text{C}=\text{O})$ in COOH), 1717 ($\nu(\text{C}=\text{O})$), 1561 ($\delta_{\text{as}}(\text{OCO})$), 1427 ($\delta_{\text{s}}(\text{OCO})$), and 1323 cm^{-1} ($\delta(\text{CH}_3)$), which are associated with the formation of acetic acid and surface acetates. These compounds can be formed owing to acetaldehyde disproportionation to acetic acid and ethanol via the Cannizzaro reaction. In addition, the formation of ethyl acetate from two acetaldehyde molecules via the Tishchenko reaction can be observed. Further heating to 200–250°C leads to the decomposition of acetates and acetic acid. In this case, the spectra exhibit bands of carbonates and bicarbonates at 1620, 1460, and $\sim 1340\text{ cm}^{-1}$. Schematic formation of the surface compounds is shown in Fig. 6.

At a low temperature, the Pd sites of the catalyst mediate the dehydrogenation of adsorbed ethanol to acetaldehyde, which in turn undergoes condensation to crotonal and C₆₊ aldehydes. In addition, acetaldehyde is converted to surface acetates and acetic acid, which, under heating, undergo decomposition to carbonates. It is significant that, under real catalysis conditions, metal sites can be blocked by substantially sorbed acetates and carbonates. Therefore, ethanol can mostly react on acid sites to form diethyl ether.

On the other hand, the IR spectroscopy data show that ethanol dehydrogenation occurs even at 50–100°C without the formation of acetates and carbonates. This finding is in agreement with the data

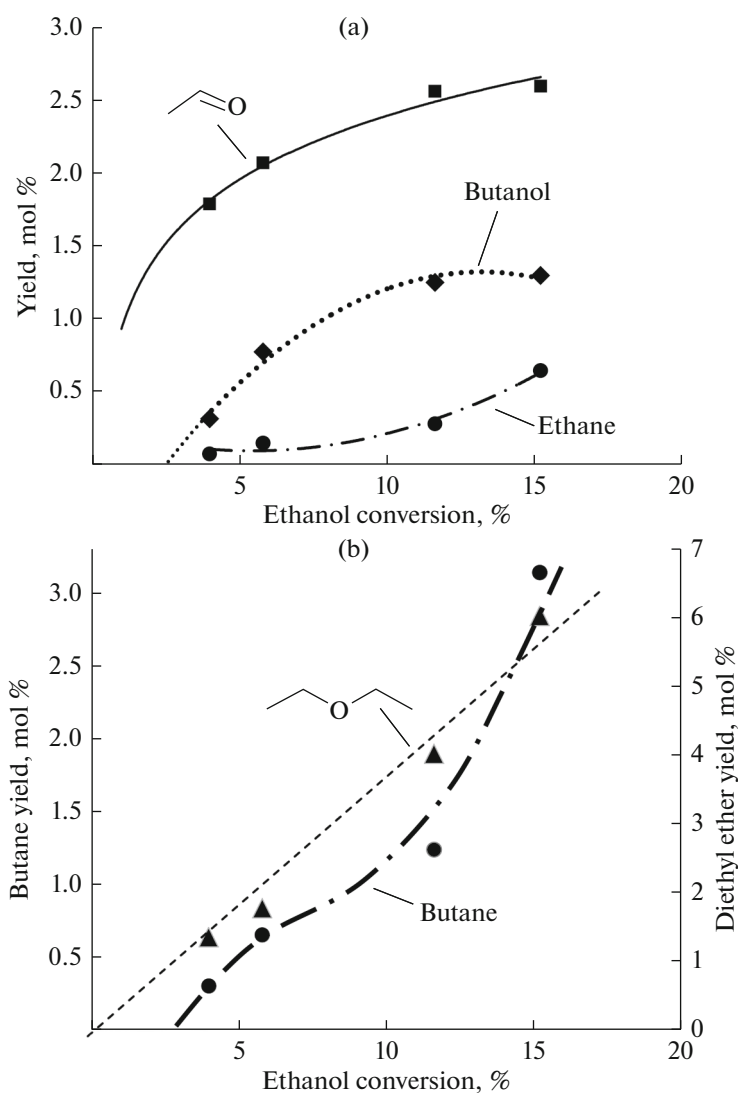


Fig. 3. Dependence of the yield of the main ethanol conversion products on conversion in the presence of the 1% Pd/Zr-BEA catalyst. Conditions: temperature, 235°C; feed space velocity, 4.3–0.7 h⁻¹; N₂/H₂/ethanol molar ratio, 2.9/1.6/1.0; and form of fed ethanol, a 50% aqueous solution.

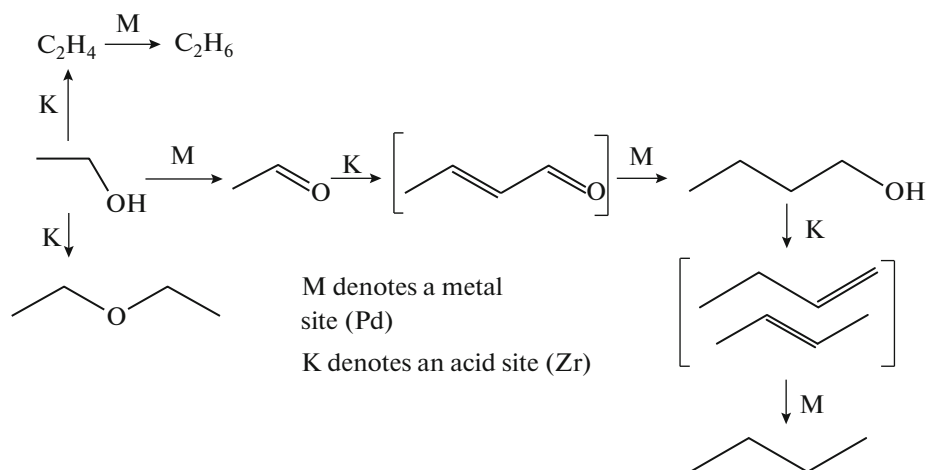


Fig. 4. Schematic formation of the main ethanol conversion products in the presence of a Pd/Zr-BEA catalyst.

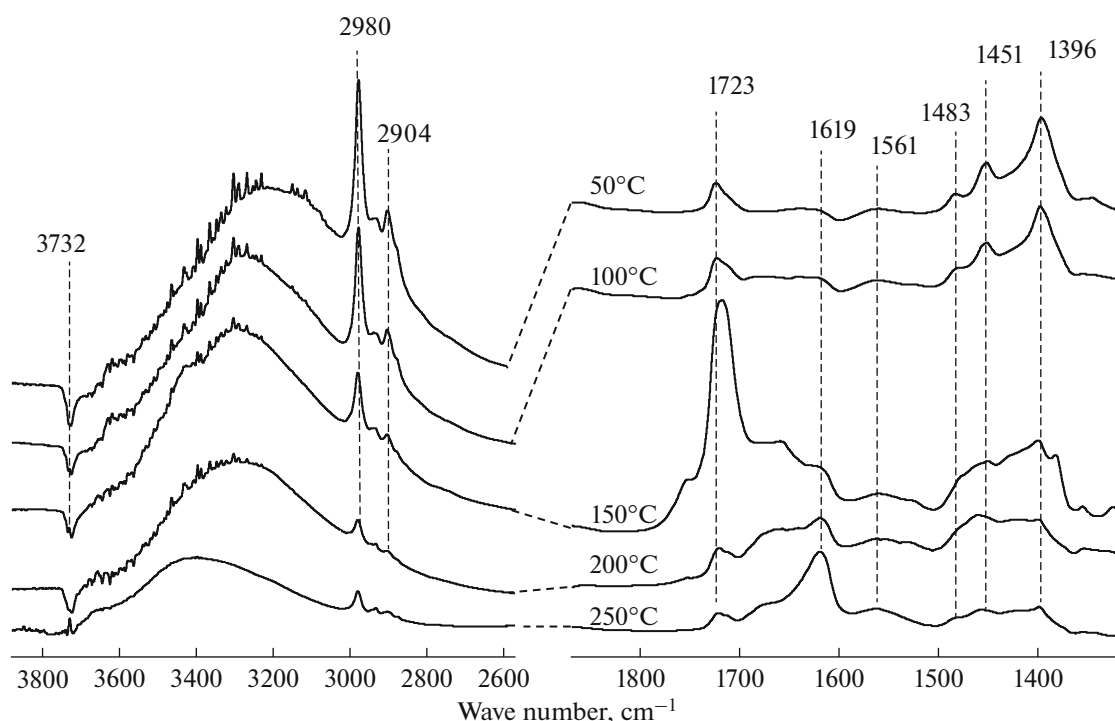


Fig. 5. Infrared spectra recorded after ethanol adsorption at 25°C and heating to 50–250°C in the presence of the 1%Pd/Zr–BEA sample.

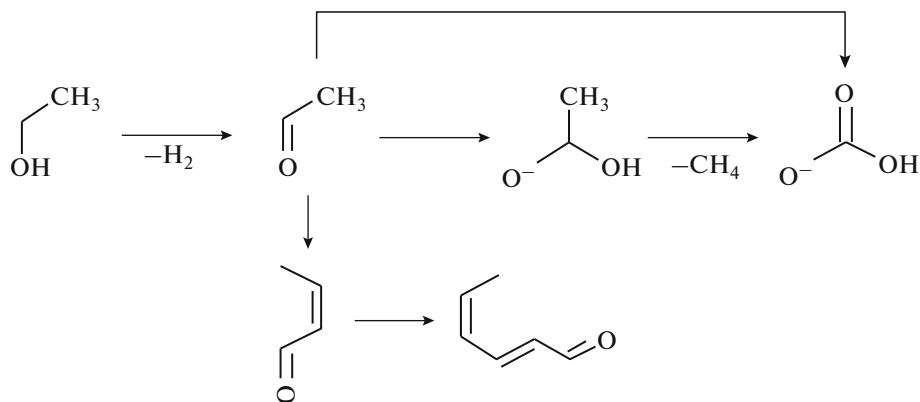


Fig. 6. Schematic formation of surface compounds according to IR spectroscopy of adsorbed ethanol.

obtained in the flow mode, where the acetaldehyde selectivity was 15–17%. Apparently, acetaldehyde undergoes condensation quite slowly; it is this factor that limits the formation of butanol over the catalyst, whereas the formation of acetaldehyde occurs quite rapidly.

Effect of Hydrogen on Ethanol Conversion

In terms of the Guerbet reaction stoichiometry, the addition of hydrogen is not necessary, because hydrogen is evolved during in situ ethanol dehydrogenation. In this case, hydrogen from the gas phase can reduce

the surface acetates and carbonates, which are readily formed from ethanol; this reduction can contribute to the purification of the Pd surface. On the other hand, hydrogen additionally fed to the system can undergo butanol hydrogenation to butane via the intermediate formation of butenes. To verify these assumptions, comparative tests with and without hydrogen feeding were conducted (Table 1).

In the case of reaction without the addition of hydrogen, the main products formed over the catalyst are acetaldehyde, diethyl ether, and butenes. Apparently, ethanol dehydrogenation occurs quite rapidly; afterwards, the resulting acetaldehyde slowly under-

goes condensation to crotonal, which, in turn, without intermediate desorption into the gas phase, is converted to butanol on the metal sites of the catalyst. The resulting butanol undergoes rigorous dehydration to form butenes.

Upon the addition of hydrogen to the stream, the diethyl ether formation rate increases twofold. In addition, the rate of formation of butanol and butane increases. The acetaldehyde formation rate decreases owing to an increase in the rate of further conversion of acetaldehyde. The amount of butenes significantly decreases, because they undergo hydrogenation to butanes. In general, the amount of C₄ hydrocarbons increases twofold; this finding suggests that hydrogen has a promoting effect on the formation of crotonal as a precursor of butanol, butenes, and butane.

Hydrogen can significantly affect the state of the active sites of the catalyst, particularly the state of the palladium particles [19]. The effect of hydrogen was studied by the ethanol-TPD method (Fig. 7).

The TPD experiment revealed that diethyl ether formed during ethanol dehydration without the addition of hydrogen to the stream is desorbed from the catalyst in the form of a broad asymmetric peak with a maximum at about 303°C; the onset of desorption is observed at 200°C. Most probably, ethanol dehydration occurs at lower temperatures; however, owing to strong binding [20], diethyl ether is desorbed from the acid site at a higher temperature. In the case of the TPD experiment using dilute hydrogen, the intensity of the peak of diethyl ether significantly decreases,

Table 1. Effect of hydrogen on the rate of formation of the main ethanol conversion reaction products

Formation rate, $\mu\text{mol}/(\text{g}_{\text{cat}} \text{ h})$	Hydrogen feeding (partial pressure of 0.29 atm)	Without hydrogen
Diethyl ether	335	169
Acetaldehyde	146	444
Butanol	100	3
Butenes	21	83
Butane	150	1
Total amount of C ₄ hydrocarbons	171	84

Conditions: catalyst, 0.6%Pd/Zr-BEA; temperature, 235°C; feed space velocity, 0.54 h⁻¹; partial pressure of ethanol, 0.18 atm; and form of fed ethanol, a 50% aqueous solution.

while the desorption maximum is shifted to lower temperatures by ~113°C. This finding is attributed to a decrease in the apparent activation energy for dehydration; a rough estimate made using the first-order reaction equation shows a decrease by 8–9 kJ/mol. The presence of hydrogen leads to the displacement of ethanol from the palladium sites; therefore, the adsorption capacity of the entire catalyst decreases. It can be assumed that, for the same reason, the chemisorbed ethanol displaced from the Pd surface moves to the acid sites, where it rapidly undergoes dehydration to diethyl ether.

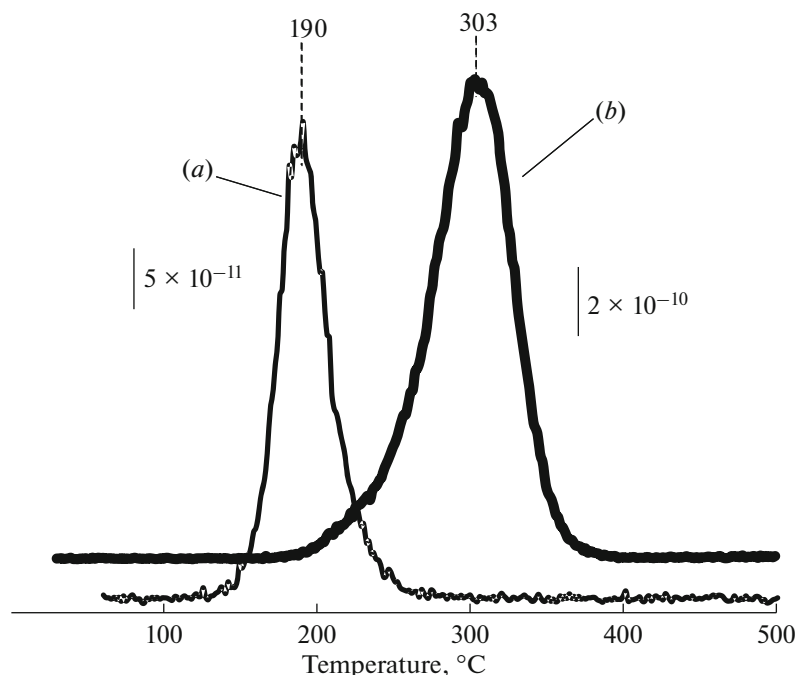


Fig. 7. Desorption of diethyl ether ($m/z = 74$) from the 0.6%Pd/Zr-BEA sample (a) in a H₂/He stream (partial pressure of hydrogen of 0.134 atm) and (b) in a pure He stream.

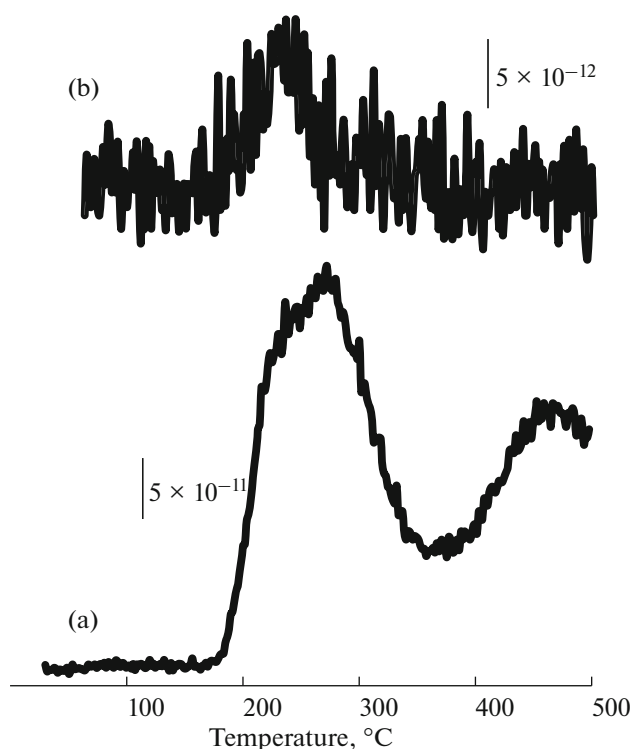


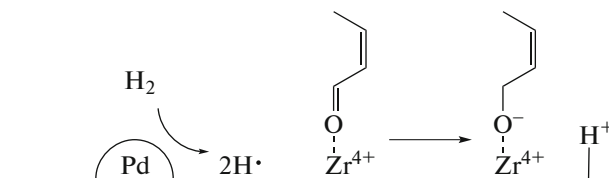
Fig. 8. Desorption of butanol ($m/z = 56$) from the 0.6%Pd/Zr-BEA sample (a) in a pure He stream and (b) in a H_2 /He stream (partial pressure of hydrogen of 0.134 atm).

Small amounts of butanol were recorded during ethanol-TPD at $m/z = 56$ (Fig. 8). According to a separate experiment with gas chromatographic analysis, an increase in temperature to $\geq 400^\circ\text{C}$ led to the formation of butenes, while butanol desorption did not occur any longer in this temperature range.

It is evident from Fig. 8 that butanol desorption occurs at a temperature of $\geq 180^\circ\text{C}$ with a maximum at 275°C . Experiments in a dilute hydrogen stream showed that the amount of desorbed butanol decreases by about 2 orders of magnitude; however, the process occurs in approximately the same temperature range. Most probably, under TPD conditions, butanol is quite rapidly converted to butane, which in turn was detected at $m/z = 43$. By analogy with ethanol conversion to diethyl ether, butanol dehydration can occur many times faster in the presence of hydrogen.

The addition of hydrogen to the reactant stream leads to a significant increase in the ethanol dehydration rate and a decrease in the apparent activation energy for this process. Therefore, the diethyl ether selectivity considerably increases. The aldol-croton condensation rate also significantly increases upon the addition of hydrogen, which is accompanied by an increase in the yield of butanol and C_4 hydrocarbons. Apparently, in the presence of hydrogen, a hydrogen spillover occurs on the surface of the Pd particles

(Scheme 2). In this case, atomic hydrogen migrates to the acid sites of the zeolite, where the adsorbed reactants and products of the acid-catalyzed stages are located [21].



Scheme 2. Formation of new acid sites on the Zr-BEA surface owing to hydrogen spillover on metal particles and subsequent hydrogenation.

The reaction of hydrogen atoms with adsorbed crotonal leads to the hydrogenation of the $C=O$ bond to form a surface enolate and a new acid site. By analogy, the hydrogenation of the $C=C$ bond in the enolate can occur. In this case, the new acid sites are localized on the siloxane bridges of the zeolite (Scheme 3).



Scheme 3. Formation of new acid sites.

The new acid sites are involved in both ethanol esterification and acetaldehyde condensation. These reactions can be accelerated owing to both an increase in the number of acid sites and the fact that the hydrogenation occurs directly on the acid site, without crotonal diffusion to the Pd particle.

Effect of Palladium Loading on Catalytic Properties

To develop a selective ethanol conversion catalyst, it is necessary to provide a balance between the acidic and hydrogenation/dehydrogenation functions of the catalyst. To this end, samples with different palladium concentrations were synthesized. The samples were compared according to the initial rates of formation of the key reaction products under low conversion conditions ($\leq 10\%$); the results are shown in Table 2.

The sample with the lowest palladium content exhibited a fairly low activity in the formation of both acetaldehyde and diethyl ether; no butanol was formed in the presence of this catalyst. For the other samples, an increase in the diethyl ether formation rate is associated with the above effect found by the ethanol-TPD method. Hydrogen spillover is capable of generating acid groups in a zeolite that mediate ethanol dehydration. The addition of 0.3% Pd to Zr-BEA leads to a fivefold increase in the diethyl ether formation rate. A further increase in the palladium content apparently leads to a change in the degree of dispersion of Pd and the coarsening of the particles. Owing to this effect, the acceleration of dehydration becomes less pronounced, because the large particles are localized separately from the acid sites of the zeolite.

Table 2. Comparison of the rates of formation of products in the presence of x% Pd/Zr–BEA samples

Sample	Product formation rate, $\mu\text{mol}/(\text{g}_{\text{cat}} \text{ h})$				Butanol selectivity, mol %
	butanol	butane	diethyl ether	acetaldehyde	
1.5 Pd	186	181	697	363	8.6
1.0 Pd	191	281	1331	510	6.3
0.3 Pd	193	311	1736	349	6.3
0.1 Pd	0	4	364	51	0.5

Conditions: feed space velocity, $1.1\text{--}1.2 \text{ h}^{-1}$; temperature, 235°C ; and feedstock, 50% ethanol/water.

Table 3. Effect of water additives on the rate of formation of products from ethanol in the presence of the 0.3% Pd/Zr–BEA catalyst

Mass fraction of water in a mixture with ethanol, %	Product formation rate, $\mu\text{mol}/(\text{g}_{\text{cat}} \text{ h})$				Butanol selectivity, mol %
	butanol	butane	diethyl ether	acetaldehyde	
5	25.4	56.9	1746.2	56.9	1.3
10	1.3	13.8	117.7	57.9	0.7
50	8.0	15.0	162.0	44.2	3.6

Conditions: $T = 235^\circ\text{C}$, WHSV = $0.4\text{--}0.8 \text{ h}^{-1}$.

The acetaldehyde formation rate increases synchronously with an increase in the palladium content in the samples. For the 1.5% Pd/Zr–BEA sample, a decrease in the rate is observed; it can be attributed to the coarsening of the particles. The butane formation rate decreases with an increase in the palladium content. Since butane is formed during the hydrogenation of butenes, which are the butanol dehydration products, the butanol formation rate directly affects the formation of butane. With allowance for this fact, it is clear that an increase in the number of Pd sites at a constant number of Zr sites leads to a decrease in the butanol dehydration rate and the rate of ethanol dehydration to diethyl ether. Comparison of butanol selectivities for catalysts with different palladium concentrations shows that the 1% Pd/Zr–BEA sample is optimal.

Effect of Water on Catalytic Properties

Ethanol conversion in the presence of Pd/Zr–BEA led to the formation of a large amount of diethyl ether. It can be assumed that the addition of water to ethanol can lead to a decrease in the dehydration reaction rate owing to the displacement of equilibrium and owing to adsorption on acid sites and their partial poisoning [22]. Results of tests with the addition of water are shown in Tables 3 and 4.

The diethyl ether formation rate decreases 14-fold upon the addition of water to the reactant stream. The butane formation rate also decreases owing to the slowing down of the dehydration of both ethanol and butanol upon the addition of water. Thermodynamic calculations show that, at 235°C , the amount of

Table 4. Comparison of experimental data with thermodynamic calculation data

Mole fraction of water in ethanol before reaction, %	Diethyl ether/ethanol molar ratio after reaction	
	thermodynamic calculation	experiment
12	2.20	0.21
22	1.95	0.05
72	0.85	0.02

Temperature, 235°C ; feed space velocity, 1.08 h^{-1} .

diethyl ether in equilibrium should be approximately an order of magnitude higher than the experimentally obtained value. Apparently, dehydration occurs in the kinetic region, and equilibrium is not established. In addition, according to calculations, an increase in the amount of water in the feed ethanol has a weaker effect on the relative content of diethyl ether than the effect found in experiments. Thus, it can be assumed that the effect of inhibition of dehydration with water is attributed to the blocking of acid sites. These sites can be localized both in the vicinity of the Pd nanoparticles and in the zeolite pores. However, it is known that Zr–BEA is a hydrophobic molecular sieve and water sorption in the pores is limited [23].

The addition of water has little effect on the apparent acetaldehyde formation rate, because the addition of water slows down the further condensation of acetaldehyde and thereby leads to a decrease in the consumption of this material for secondary products.

Since the apparent acetaldehyde formation rate remains unchanged, it can be assumed that the ethanol dehydrogenation rate decreases with an increase in the amount of water. The butanol formation rate also decreases; this fact suggests that acetaldehyde condensation is slowed down. Thus, the addition of water inhibits all acid-catalyzed stages of the process and slows down the dehydrogenation and hydrogenation reactions over palladium particles.

CONCLUSIONS

The main products of ethanol conversion in the presence of a Pd/Zr–BEA catalyst have been determined; the order of formation of the products has been revealed. Ethanol is converted to butanol via the intermediate formation of acetaldehyde. The stage of ethanol dehydrogenation to acetaldehyde on Pd sites is fairly rapid; the rate of the entire process is limited to the aldol–croton condensation of acetaldehyde to crotonal. The main side processes are ethanol dehydration to diethyl ether and butanol to form butenes, which undergo hydrogenation to form butane. At higher temperatures, acetaldehyde is converted to acetates and carbonates on the catalyst surface, which can block the metal sites of the catalyst. In the presence of hydrogen in the gas phase, the ethanol dehydration rate, along with the rate of the aldol–croton condensation of acetaldehyde, significantly increases. This effect is attributed to hydrogen spillover to the zeolite and the formation of new acid sites in the vicinity of the Pd particles. It has been found that the addition of water to the ethanol stream slows down the dehydration owing to the blocking of the acid sites with water.

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